A point estimate needs a loss function

Posterior mean Slide 63

$$p(z \mid y) = \mathcal{N}\big(z \mid \mu_{z|y}, \Sigma_{z|y}\big), \quad E[z \mid y] = \mu_{z|y}$$

$$y \text{ observed}, \quad a \in \mathbb{R}^d, \quad E[\|z\|^2 \mid y] < \infty$$

$$\ell(a, z) = \|z - a\|^2, \quad R(a \mid y) = E[\ell(a, z) \mid y]$$

$$\begin{aligned} \|z - a\|^2 &= \big\|\big(z - \mu_{z|y}\big) + \big(\mu_{z|y} - a\big)\big\|^2 \\ &= \big\|z - \mu_{z|y}\big\|^2 + 2\big(z - \mu_{z|y}\big)^T\big(\mu_{z|y} - a\big) + \big\|\mu_{z|y} - a\big\|^2 \end{aligned}$$

- A point estimate is judged by its posterior expected loss.